

Overview of Adversarial machine learning

Subject: Adversarial machine learning

1.

Adversarial machine learning is the study of the attacks on machine learning algorithms, and of the defenses against such attacks. Machine learning techniques are mostly designed to work on specific problem sets, under the assumption that the training and test data are generated from the same statistical distribution (IID). However, this assumption is often dangerously violated in practical high-stake applications, where users may intentionally supply fabricated data that violates the statistical assumption. Most common attacks in adversarial machine learning include evasion attacks, data poisoning attacks, Byzantine attacks and model extraction.

2.

Untargeted: $\min_{x'} d(x', x) \text{ subject to } C(x') \neq C(x)$ $\{\text{displaystyle } \textbf{Untargeted:} \} \min_{x'} d(x', x) \text{ subject to } C(x') \neq C(x)$

3.

adversarial training of a linear regression model with input perturbations restricted by the infinity-norm closely resembles Lasso regression, and that adversarial training of a linear regression model with input perturbations restricted by the 2-norm closely resembles Ridge regression.

4.

$\nabla S(x', \delta) \approx \frac{1}{B} \sum_{b=1}^B \nabla \phi(x' + \delta u_b) u_b$ $\{\text{displaystyle } \nabla S(x', \delta) \approx \frac{1}{B} \sum_{b=1}^B \phi(x' + \delta u_b) u_b \}$

5.

Adversarial examples An adversarial example refers to specially crafted input that is designed to look "normal" to humans but causes misclassification to a machine learning model. Often, a form of specially designed "noise" is used to elicit the misclassifications. Below are some current techniques for generating adversarial examples in the literature (by no means an exhaustive list).

6.

Byzantine attacks As machine learning is scaled, it often relies on multiple computing machines. In federated learning, for instance, edge devices collaborate with a central server, typically by sending gradients or model parameters. However, some of these devices may deviate from their expected behavior, e.g. to harm the central server's model or to bias algorithms towards certain behaviors (e.g., amplifying the recommendation of disinformation content). On the other hand, if the training is performed on a single machine, then the model is very vulnerable to a failure of the machine, or an attack on the machine; the machine is a single point of failure. In fact, the machine owner may themselves insert provably undetectable backdoors. The current leading solutions to make (distributed) learning algorithms provably resilient to a minority of malicious (a.k.a. Byzantine) participants are based on robust gradient aggregation rules. The robust aggregation rules do not always work especially when the data across participants has a non-iid distribution. Nevertheless, in the context of

heterogeneous honest participants, such as users with different consumption habits for recommendation algorithms or writing styles for language models, there are provable impossibility theorems on what any robust learning algorithm can guarantee.

7.

Evasion Evasion attacks consist of exploiting the imperfection of a trained model. For instance, spammers and hackers often attempt to evade detection by obfuscating the content of spam emails and malware. Samples are modified to evade detection; that is, to be classified as legitimate. This does not involve influence over the training data. A clear example of evasion is image-based spam in which the spam content is embedded within an attached image to evade textual analysis by anti-spam filters. Another example of evasion is given by spoofing attacks against biometric verification systems. Evasion attacks can be generally split into two different categories: black box attacks and white box attacks.

8.

Performance evaluation The choice of the evaluation metrics can impact the validity of the results. Having an inappropriate baseline involves failing to compare a new model against simpler, well-established baselines. Inappropriate performance measures means using metrics that do not align with the practical goals of the system. Reporting only "accuracy" is often described as insufficient for an intrusion detection system, where false-positive rates are considered critically important. Base rate fallacy is a failure to correctly interpret performance in the context of large class imbalances.

9.

$$\min_{\hat{x} \in [0, 1]^d} L(f(\hat{x}), y), \text{ s.t. } \|\hat{x} - x\|_p \leq \epsilon$$

10.

Feature robustness Researchers differentiate between features that can be easily manipulated and those that are more resistant to modification. For example, simple static attributes, such as header fields, may be altered by attackers, while structural features, such as control-flow graphs, are generally more stable but computationally expensive to extract.

11.

Labeling and ground truth challenges Malware labels are often unstable because different antivirus engines may classify the same sample in conflicting ways. Ceschin et al. note that families may be renamed or reorganized over time, causing further discrepancies in ground truth and reducing the reliability of benchmarks.

12.

Secure learning algorithms Byzantine-resilient algorithms Multiple classifier systems AI-written algorithms. AIs that explore the training environment; for example, in image recognition, actively navigating a 3D environment rather than passively scanning a fixed set of 2D images. Privacy-preserving learning Ladder algorithm for Kaggle-style competitions Game theoretic models Sanitizing training data Adversarial training Backdoor detection algorithms Gradient

masking/obfuscation techniques: to prevent the adversary exploiting the gradient in white-box attacks. This family of defenses is deemed unreliable as these models are still vulnerable to black-box attacks or can be circumvented in other ways. Ensembles of models have been proposed in literature, which have shown to be ineffective against evasion attacks but effective against data poisoning attacks. Moving target defense

13.

$\min (|| \delta ||_p)$ subject to $f (x + \delta) \leq 0$, $x + \delta \in [0 , 1]^n$ $\{\displaystyle \min(||\delta ||_p)\}$ $\{\text{subject to }\}$ $f(x+\delta)\leq 0, x+\delta \in [0,1]^n\}$

14.

Data collection issues Security datasets vary across formats, including binaries, network traces, and log files. Studies have reported that the process of converting these sources into features can introduce bias or inconsistencies. In addition, time-based leakage can occur when related malware samples are not properly separated across training and testing splits, which may lead to overly optimistic results.